1. Competency #2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

(Judge whether a mathematical result is reasonable within the problem’s real-world context)

- Highly relevant.
  Students must calculate that the current available space is 152 MB, the required additional space is 350 MB, and thus they need to free at least an extra 198 MB. They then need to select up to two music albums whose combined size is ≥198 MB.
- During problem solving, if a student concludes that deleting Album 4 (55 MB) and Album 5 (60 MB)—totaling 115 MB—is sufficient, the arithmetic may be correct, but the conclusion fails to meet the actual requirement (115 < 198). This shows a failure to assess whether the mathematical outcome makes sense in context.
- Therefore, students must verify that their chosen solution truly satisfies the real-world constraint of freeing at least 198 MB, demonstrating their ability to judge the contextual reasonableness of results.

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2. Competency #6: Use knowledge of how the real world impacts the outcomes and calculations… to make contextual judgments about how the results should be adjusted or applied

(Apply real-world knowledge to adjust or apply mathematical results appropriately)

- Highly relevant.
  The problem imposes explicit real-world constraints:
  - Photos cannot be deleted;
  - At most two music albums may be deleted;
  - Storage is measured in MB, with a total capacity of 1 GB (= 1000 MB).
- These are not purely mathematical rules but reflect practical limitations of digital device usage. Students must incorporate these non-mathematical factors into their decision-making—for example, even if mathematically feasible to delete three small albums to meet the space requirement, the problem explicitly forbids doing so.
- Thus, solving the problem involves not just combinatorial optimization but also respecting and applying real-world constraints.

3. Competency #3: Identify the limits of the model used to solve a problem

(Recognize the limitations of the mathematical model employed)

- Moderately relevant.
  The implicit “model” here assumes that storage space is perfectly additive: total space = used + free, and file sizes can be summed and subtracted without loss.
- Although the problem does not require students to critically reflect on model limitations (e.g., file system overhead, storage fragmentation, or compression), more advanced students might recognize that in reality, a labeled 1000 MB capacity ≠ fully usable space, or that deleted files may not immediately release space.
- While not deeply assessed, awareness of these simplifying assumptions constitutes a basic manifestation of this competency.

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4. Competency #1: Interpret a mathematical result back into the real-world context

(Translate mathematical results back into meaningful real-world actions or conclusions)

- Moderately relevant.
  After calculating that deleting Album 1 (100 MB) and Album 8 (125 MB) frees 225 MB, the student must interpret this as: “This yields 152 + 225 = 377 MB > 350 MB, which is sufficient for the new photo album.”
- This step—converting an abstract number (225 MB) into a concrete action (which albums to delete) and a practical outcome (successful storage)—is a classic example of interpretation (or “recontextualization”) of mathematical results.
- Though straightforward, it clearly engages this competency.

5. Competency #5: Interpret mathematical outcomes in a variety of formats… compare representations

(Understand results across different representational formats and compare them)

- Weakly relevant.
  The problem provides information in two forms: a textual "pie-chart-like" breakdown (Music: 650 MB, Photos: 198 MB, Free: 152 MB) and a table listing individual album sizes. Students must integrate both sources.
- However, the task does not ask students to compare the effectiveness of different representations or convert between formats, so the connection to this competency is limited.

1. 第2项：Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

（判断数学结果或结论在问题情境中是否合理）

- 高度相关。
  学生需计算当前可用空间（152 MB），目标新增空间需求（350 MB），差额为 198 MB。接着需从现有音乐专辑中选出至多两个，其总大小 ≥198 MB。
- 解题过程中，若学生得出“删除 Album 4（55 MB）和 Album 5（60 MB）共 115 MB”就足够，那这个数学计算虽正确，但不符合实际需求（115 < 198），说明未结合上下文判断结果合理性。
- 因此，学生必须验证所选方案是否真正满足“腾出 ≥198 MB”的现实约束，体现对结果在情境中合理性的判断。

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2. 第6项：Use knowledge of how the real world impacts the outcomes and calculations… to make contextual judgments about how the results should be adjusted or applied

（利用现实世界知识对数学结果进行情境化调整或应用）

- 高度相关。
  题目设定了明确的现实约束：
  - 不能删除照片；
  - 最多只能删两个音乐专辑；
  - 存储单位以 MB 计，总量有限（1 GB = 1000 MB）。
- 这些都不是纯数学规则，而是现实设备使用中的限制条件。学生必须将这些非数学因素纳入决策过程，例如即使数学上可以删三个小专辑凑够空间，但题目禁止这样做。
- 因此，解题不仅是组合优化，更是对现实约束的尊重与应用。

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3. 第3项：Identify the limits of the model used to solve a problem

（识别所用模型的局限性）

- 中等相关。
  本题隐含的“模型”是：存储空间 = 已用 + 剩余，且文件大小可简单加减。
- 虽然题目未要求学生反思模型局限（如：是否考虑文件系统开销、碎片化、压缩等），但有能力的学生可能会意识到：现实中 1000 MB 标称容量 ≠ 实际可用容量，或删除文件后空间是否立即释放等。
- 尽管题目未深入考察这点，但理解该简化模型的假设（即空间完全可加、无损耗）属于此项能力的初级体现。

4. 第1项：Interpret a mathematical result back into the real world context

（将数学结果还原到现实情境中）

- 中等相关。
  学生计算出“删除 Album 1（100 MB）和 Album 8（125 MB）共 225 MB”后，需将其解释为：“这样就能腾出 152 + 225 = 377 MB > 350 MB，足够存新相册”。
- 这一过程就是将抽象数字（225）转化为具体行动（删哪两个专辑）和现实结果（能否存入），属于典型的“回译”（interpretation）。
- 虽然操作较直接，但仍体现了此项能力。

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5. 第5项：Interpret mathematical outcomes in a variety of formats… compare representations

（以多种格式解读数学结果，并比较表示方式）

- 弱相关。
  题目提供了饼图式文字描述（Music 650MB, Photos 198MB, Free 152MB）和表格（专辑大小）。学生需整合两种信息源。
- 但并未要求比较不同表示法的优劣，或转换图表形式，因此此项关联较弱。